BROADBAND NETWORKS - PRACTICAL PROJECT

Simulating a 5G Network with OMNeT++ and Simu5G

A hands-on walkthrough: building a simulation stack and running three experiments - from the 1970s ALOHA protocol to a small 5G cell.

OMNeT++ 6.1 INET 4.5 Simu5G 1.3

OVERVIEW

What we did

- Installed and built a complete network-simulation stack (OMNeT++, INET and Simu5G) from scratch on Linux.
- Ran three experiments of growing complexity: a shared radio channel, a wired LAN, and a small 5G network.
- Measured the real performance of each one: collisions, throughput, delay and radio signal quality.
- Everything runs live in a graphical interface, so each part can be shown step by step.

THE TOOLS

Three programs, stacked on top of each other

Simu5G — Adds 5G on top: base stations, phones, the radio channel and scheduler.

INET — A big library of ready-made networks: Ethernet, IP, Wi-Fi, routers, hosts.

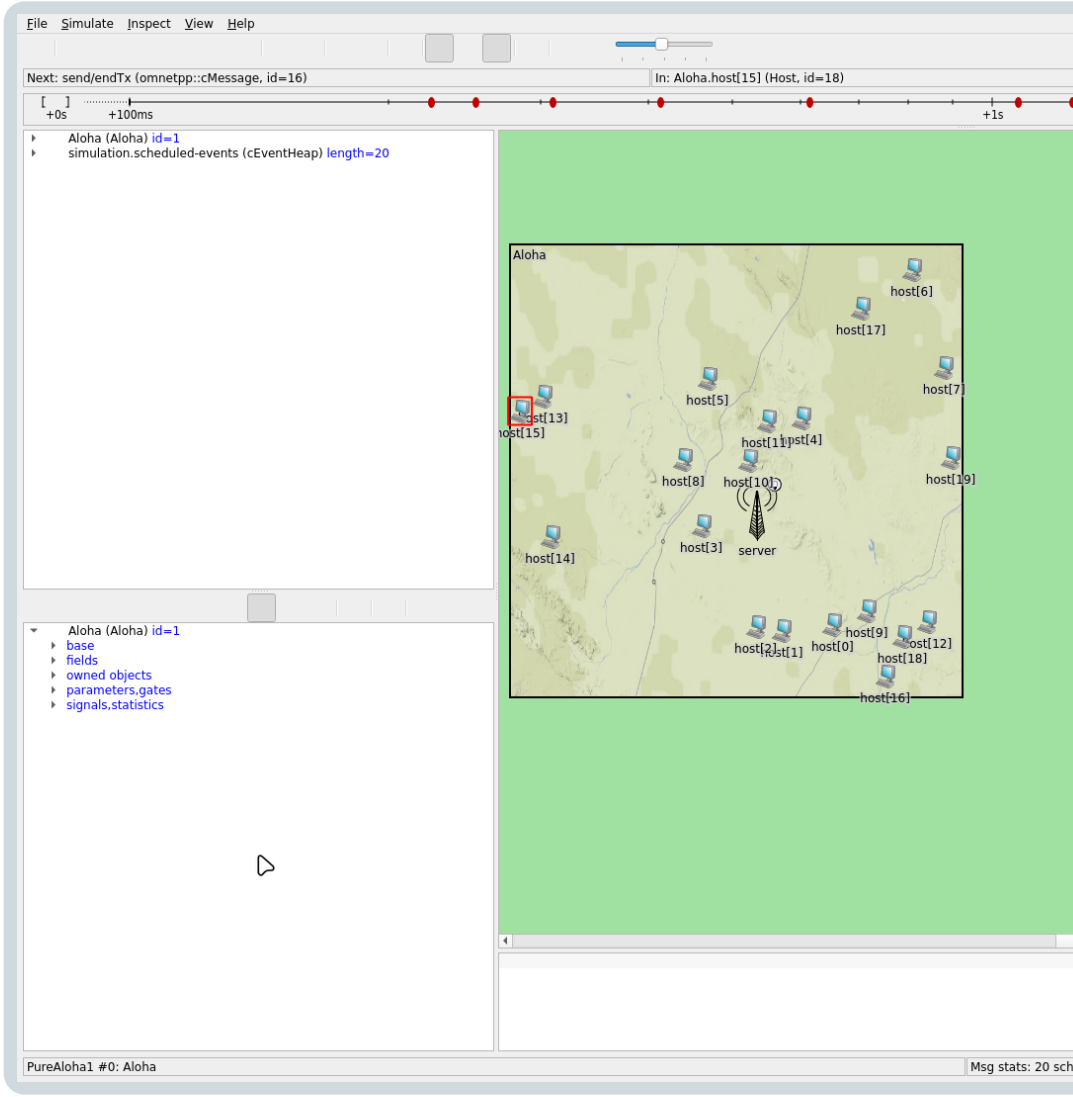
OMNeT++ — The engine underneath everything: it runs the simulation and draws the GUI.

You always run them together - each layer needs the one below it.

TASK 1 - ALOHA

Sharing one radio channel

- ALOHA is the oldest way to share a channel: every station transmits whenever it wants.
- If two transmit at the same time, the signals collide and both messages are lost.
- We ran it under light, optimal and heavy traffic to see the trade-off.



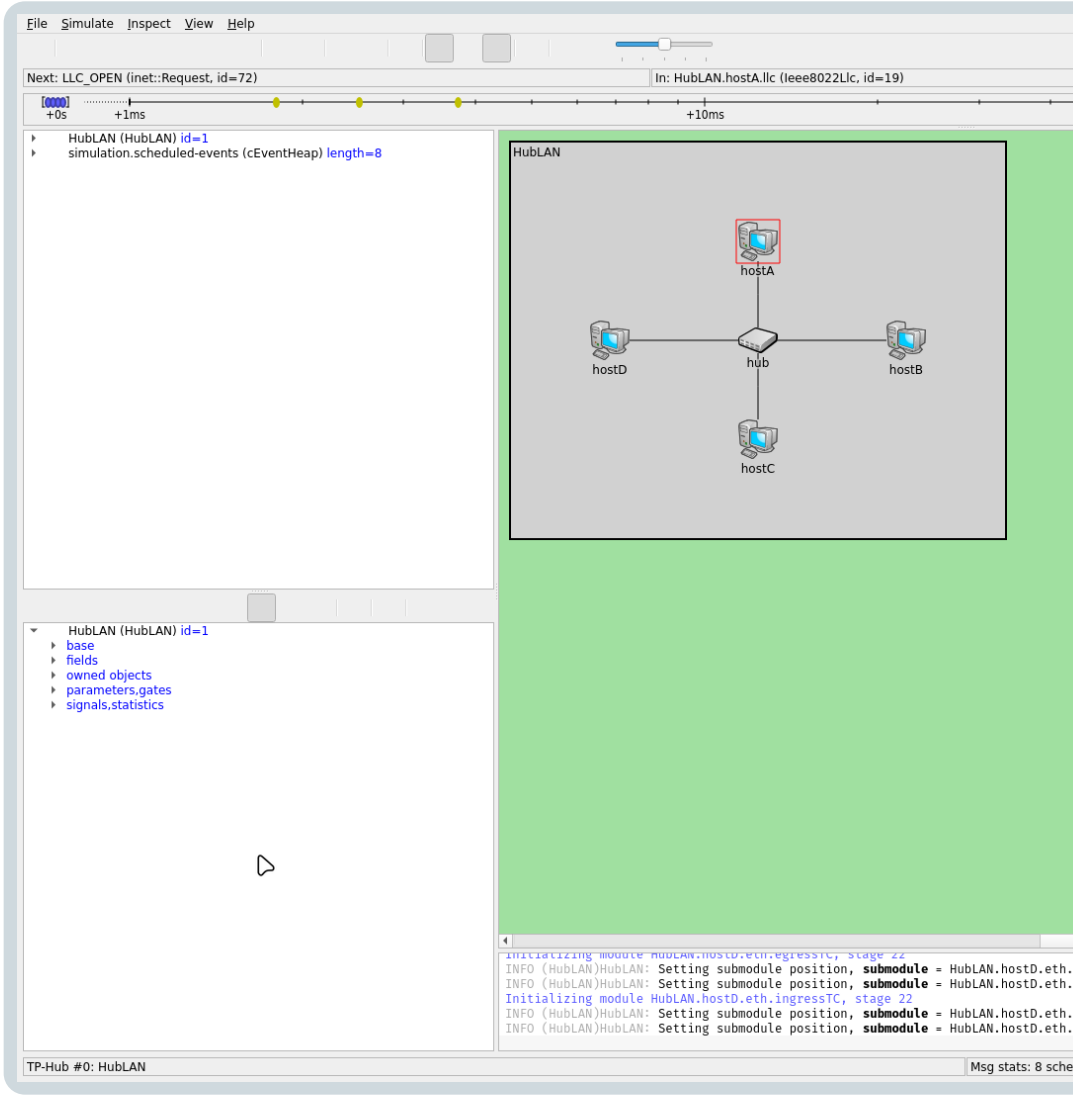
RESULT

Best case reached 17.3% channel use - right at the theoretical limit of ~18%. Too much traffic just makes collisions waste the channel.

TASK 2 - WIRED ETHERNET

A hub versus a switch

- A hub shares one wire between all devices, so they collide - just like ALOHA.
- A switch gives each device its own line, so collisions disappear.
- We sent the same heavy traffic to both, and built a custom 6-host network too.



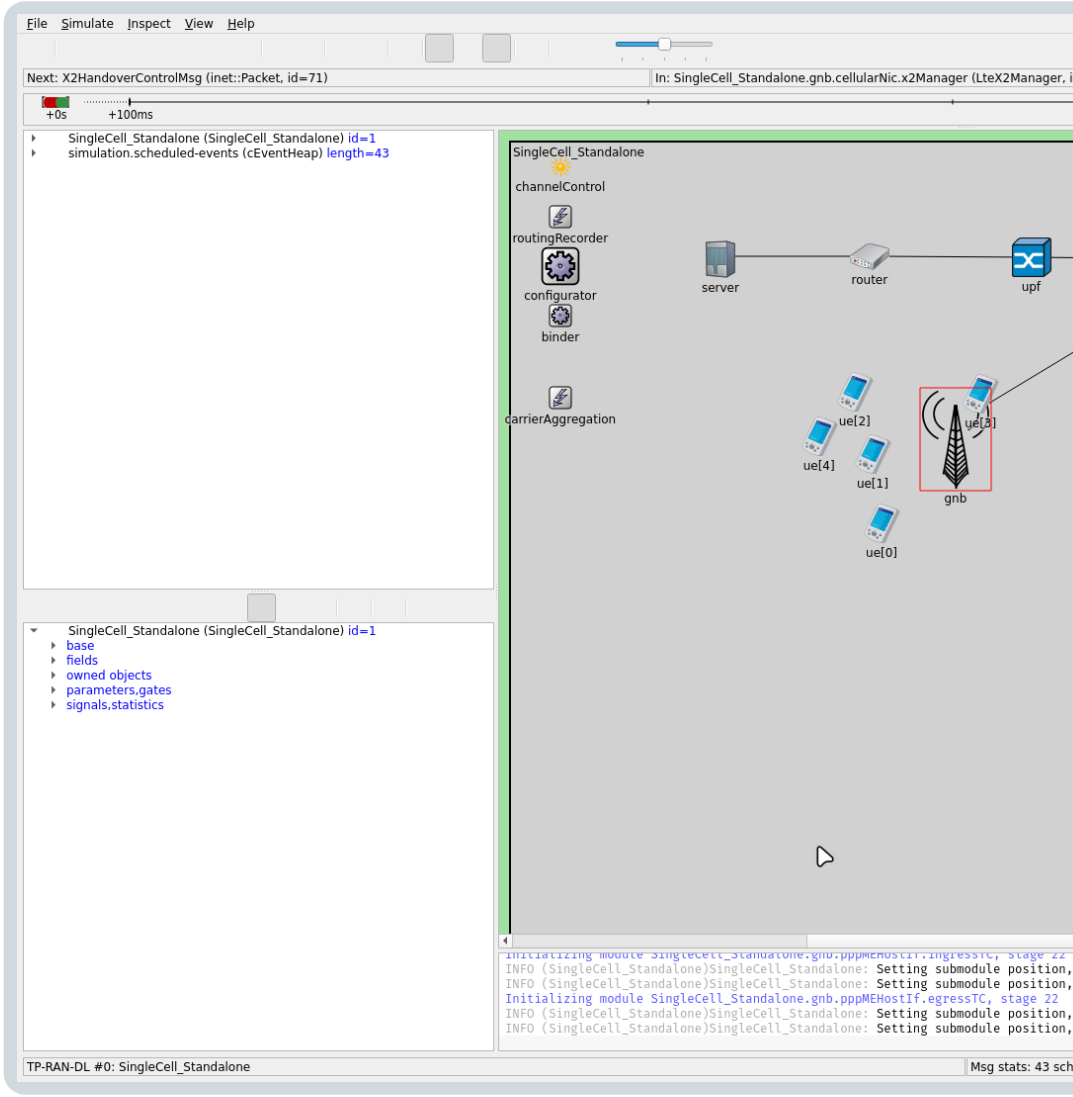
RESULT

Same load: the hub had 172,647 collisions (24% of the time wasted); the switch had 0. Our 6-host hub rose to 500,245 (45%).

TASK 3 - 5G NETWORK

One base station, five phones

- We built a small 5G cell: one base station (gNB) serving five moving phones (UEs).
- A server on the Internet sends each phone a steady stream of data over 5G.
- We measured what a real operator cares about: speed, delay, loss and signal quality.



RESULT

Every phone received its full ~0.8 Mb/s with zero loss, 4-5 ms delay, and a near-perfect radio signal (CQI ~15 out of 15).

RESULTS AT A GLANCE

The numbers that came out

17.3%

best ALOHA channel use (theory limit ~18%)

0

collisions once we used a switch

5 / 5

phones served in 5G with zero packet loss

4-5 ms

end-to-end delay across the 5G cell

TAKEAWAYS

What this shows

- The whole stack works end to end - from a 1970s protocol all the way to a working 5G cell.
- Sharing one medium causes collisions; switches and the 5G scheduler are what remove them.
- The 5G network delivered every user its full data rate, with low delay and no loss.

All three run live in the GUI with: bash ~/omnetpp-ws/demo.sh